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Assignment 1: observations on infection.txt

Running the C code found in the translated infection.txt on a VM running Linux Mint caused the computer to, essentially, lose all functionality, with running programs freezing and any new programs being prevented from being launched. Attempting to start new programs created an error for “failed to fork (resource temporarily unavailable). This even prevented me from logging out or even shutting the machine down via the taskbar menu.

Running the same code on a Windows 11 machine initially appeared to yield less severe results. During execution, programs could still be run and there was no noticeable lag; the execution was still able to be halted by CTRL+C in the command prompt window. The only obvious symptom while execution occurred was in Task Manager, where the user interface completely went haywire and the program stopped responding to any input. Task Manager remained nonfunctional until the machine was restarted, and attempting to use it before then caused severe lag. However, about half an hour later, similar symptoms of programs freezing and new programs being unable to start began to suddenly occur, forcing me to restart the computer.

Given these observations, I believe this code is a ‘fork bomb’; it repeatedly opens copies of itself and floods the system’s memory. This may be useful for attackers as a crude form of denial-of-service attack, causing a machine to be effectively crippled by preventing process execution. This likely leaves the system in a vulnerable state where it may be susceptible to attack, or it may also force the system to be restarted, allowing a brief window without protection. One possible method to counter such an attack would be to detect how much memory a program is using and to forcibly restrict that amount if the demand is unreasonably high; another more preemptive method would be preventing the intrusion of such a file onto the device to begin with through other means.